SIFFI SINGH

siffisingh26@gmail.com | +1 626.298.9751 | Los Angeles, US | linkedin.com/in/siffi | siffi26.github.io

PROFESSIONAL SUMMARY

Applied Scientist with 7 years delivering production AI systems at Amazon AWS. Known for translating cutting-edge research into high-impact customer solutions, from Generative AI-powered trend detection serving thousands of enterprise customers, to LLM-based pipelines deployed across 14 languages. Combines deep technical rigor with a strong bias for action and customer-backward thinking. Recognized by managers and peers for technical excellence, cross-functional leadership, and a proven ability to mentor and elevate teams. Published at top NLP venues ACL, NAACL, EMNLP, and Best Paper Award at IEEE ICCE.

EXPERIENCE

Amazon Development Center Sep 2019 – Present

Applied Scientist II, Amazon Web Services Jul 2023 – Present | **Los Angeles, US**

- Automatic Speech Translation: Call centers and healthcare services struggled with slow, error-prone translation pipelines that made real-time multilingual communication impractical. Built an end-to-end proof-of-concept for AWS Transcribe that takes audio and outputs translated text directly. Demonstrated bidirectional English-Spanish translation with a high BLEU score of 70%, comparable to trained ASR and LLM-based systems.
- Generative AI Trend Detection**:** Businesses could identify themes in a snapshot but had no way to track how customer issues evolve and emerge over months of conversations. Led research and productionization of AWS Automated Trend Detection in Amazon Connect, a Generative AI pipeline that monitors themes longitudinally, surfacing rising and recurring issues over time. Launched in private beta with enterprise customers including Centrica and Air Canada, with multilingual support for French and Spanish.

Applied Scientist, Amazon Web Services Sep 2021 – Jun 2023 | **Cambridge, UK**

- PII Redaction: Global AWS customers lacked reliable privacy protection for sensitive data across multilingual call and chat interactions. Led expansion of AWS Transcribe's PII redaction system, leveraging LLM-generated synthetic data to retrain XLM-RoBERTa with language-optimized tokenization. Scaled to 14 languages including European (PFIGS), Asian (CJK, Cantonese, Taiwanese) and 6 English locale variants, protecting bank account and credit card data for thousands of global AWS customers.
- Theme Detection (Contact Lens)**:** Businesses needed to understand what customers were calling about across thousands of call and chat conversations but manually labeling them was infeasible. Designed an automated ML pipeline using semantic embeddings, clustering, and a BART-based model to identify and label themes. Achieved 88% on human evaluation for Cluster Coherence and Label Quality; deployed to 4000+ enterprise customers including Barclays and Air Canada.
- Semantic Matching: Contact Lens customers searching thousands of conversations by keyword missed results where meaning didn't match exact words. Built and evaluated embedding models and pooling strategies, deploying a single unified model that understands intent rather than keywords. Achieved 70% average precision across both call and chat domains.

Software Development Engineer, Amazon Alexa Sep 2019 – Sep 2021 | **Cambridge, UK**

- Relation Extraction from Web Tables**:** Alexa's knowledge graph couldn't scale without automated fact extraction from the web. Trained SUREAL, a semi-supervised relation alignment system, to predict relations between web table columns and align them to Alexa's Knowledge Graph. Benchmarked against TaBERT on the T2Dv2 golden dataset. Published at Amazon Machine Learning Conference (AMLC) 2020. Built an end-to-end service to ingest structured facts from tabular data into Alexa knowledge graphs.

- RedTab - Tabular Benchmark Dataset: No domain-specific benchmark existed for column class detection and relation extraction from web tables. Led end-to-end creation of RedTab, 5K annotated tables across music and literature domains, covering ontology design, crowdsourced annotation, and quality evaluation. Published at COLING 2022 and used as the production benchmark for Alexa's relation extraction models.
- Knowledge Graph Ingestion & Maintenance: Alexa's knowledge graph lacked entertainment coverage and contained stale facts that degraded answer quality. Built an end-to-end pipeline to extract structured facts from tabular web data and ingest them into Alexa's Evi Knowledge Graph. Ingested 11M facts across IMDB Movies and People, growing the graph by +130% (44.5M to 103M entities). Removed stale facts to keep Alexa's answers accurate.

California Institute of Technology Dec 2025 – Mar 2025 | Los Angeles, US

External Collaborator with Prof. Sergei Gukov Division of Physics, Mathematics and Astronomy

- Andrews-Curtis Conjecture**:** Applying greedy search, breadth-first search, and a PPO reinforcement learning agent to solve hard presentations from a long-standing open conjecture in combinatorial group theory. Solved 630 out of 1190 presentations from the Miller–Schupp series, surpassing prior state-of-the-art. Exploring improved reward structures and agent environments to extend results to 3-generator families and apply RL to formally prove mathematical conjectures.

University of Cambridge Oct 2024 – Jan 2025 | Cambridge, UK

External Collaborator with Prof. Burigede Liu Department of Engineering

- Material Property Prediction: Materials databases have significant missing values, making material selection costly and slow. Built an automated pipeline to extract and process 17,052 metal materials from MatWeb, implementing a BERT-based model to predict missing properties by learning interdependencies between physical properties. Achieved a prediction error of 2.98 MSE within ±10% tolerance**,** with LLMs matching trained model performance using just a few examples suggesting a scalable path to materials discovery without expensive experimentation.

JMicron Technology Jun 2017 – Jun 2019 | Hsinchu, Taiwan

Research Intern / Thesis with Prof. Hsueh-Ming Hang, NCTU

- Face Clustering & Recognition: Consumer devices lacked accurate, privacy-preserving face recognition due to limited hardware and real-world challenges such as varying illumination, pose, and age. Designed a lightweight unsupervised deep learning system in TensorFlow, optimized for constrained hardware, that automatically identifies, and groups faces in personal photo collections. Applied a progressive multi-stage training approach to handle real-world face variations. Achieved 86% average ARI accuracy, commercially launched as product ‘LivCore’. Our paper "A Continuous Learning System for Face Clustering and Recognition" received Best Paper Award in IEEE ICCE 2021.

Industrial Technology Research Institute Jun 2018 – Aug 2018 | Hsinchu, Taiwan

Research Intern

- Autonomous Vehicle Perception: Taiwan's first intelligent computer vision team for autonomous vehicles lacked standardised benchmark data for evaluating perception algorithms. Developed a benchmark dataset and evaluated Image and LiDAR fusion data for segmentation and tracking algorithms using Robot Operating System (ROS)**,** enabling reliable performance measurement for autonomous vehicle systems.

National Chiao Tung University Feb 2017 – Jun 2017 | Hsinchu, Taiwan

Research Intern

- Autonomous Vehicle Classification: DSP-based autonomous car systems required a lightweight, real-time classification algorithm robust to sensor noise. Implemented a fixed-point C MoDSeT classification algorithm using a 3-D millimeter wave radar sensor on a DSP processor, ensuring system robustness and real-time performance.

EDUCATION

Master of Science – Electrical and Computer Science Engineering

National Chiao Tung University — Jun 2017 – Jun 2019 | Hsinchu, Taiwan

Supervisor: Prof. Hsueh-Ming Hang | GPA: 4.0/4.0

Thesis: Deep Learning Based Face Clustering and Recognition System - Designed a lightweight unsupervised deep learning system to automatically identify and group faces in personal photo collections, optimized for devices with constrained hardware.

Relevant Coursework: Machine Learning, Deep Learning, Algorithms, Artificial Intelligence, Computer Vision, User Experience and Usability, Source Coding, Service-Oriented Architecture

Bachelor of Technology – Computer Science and Engineering

Dr. APJ Abdul Kalam Technical University — Aug 2013 – Jun 2017 | Noida, India

Supervisor: Prof. Carrson C. Fung | Ranked 1st out of 10,000+ students | GPA: 3.8/4.0 | First Division with Distinction

Relevant Coursework: Data Structures, Algorithms, Distributed Systems, Neural Networks, Pattern Recognition, Cryptography, Computer Networks, Digital Image Processing, Software Engineering, Compiler Design, Discrete Structures & Graph Theory

SKILLS

- Programming Languages**:** Python, Java, Scala, C, C++, MATLAB
- ML & Data Frameworks: PyTorch, TensorFlow, HuggingFace Transformers, Sentence-Transformers, PySpark, Scikit-learn, OpenCV, CUDA, NLTK, SpaCy, UMAP, NumPy, Pandas, Matplotlib, NetworkX
- Cloud & AWS: Transcribe, Comprehend, Connect, SageMaker, Lex, Lambda, EC2, EMR, S3, DynamoDB, SNS, SQS, CloudFormation, Glue, SWF, Step Functions
- AI, NLP & Research: Large Language Models (LLMs), Generative AI, Natural Language Processing, Conversational AI, Speech Recognition, Multilingual NLP, Computer Vision, Deep Neural Networks, Deep Reinforcement Learning
- Research & Engineering: Algorithms, Data Structures, Systems Design, Distributed Systems, Automation

HONORS & AWARDS

- Best Paper Award, IEEE International Conference on Consumer Electronics (ICCE) 2021 for research on continuous learning systems for face clustering and recognition, commercially launched as LivCore.
- Organized Track 2 at Dialog System Technology Challenge (DSTC) 12 - Led the Controllable Conversational Theme Detection track, attracting 10 teams; findings published in DSTC 12 proceedings.
- Silver Medal, ACM ICPC Asia Jakarta Regionals 2017 - one of the world's most prestigious collegiate coding competitions; previously qualified at ACM ICPC India Amritapuri Regionals 2016.
- 1st Place, Amazon Alexa Hackathon 2020 **-** Built *QUAC (Quite Useful Assistant for Coding)*, a debugging chatbot using Amazon Lex that guides developers through structured problem-solving, searches Stack Overflow, and matches teammates with similar issues.
- Academic Excellence Scholarship - Awarded by Dr. APJ Abdul Kalam Technical University for undergraduate research internship, and independently by National Chiao Tung University (NCTU) for graduate studies in Electrical Engineering and Computer Science, Taiwan.
- Google Women Techmakers Scholarship 2018 - Attended Google Chrome Dev Summit, San Francisco.
- Founded University's Programming Club - Established the first programming club at Dr. APJ Abdul Kalam Technical University, growing a community of 100+ students and building a culture of competitive programming and student-led hackathons.

PUBLICATIONS

[1] HP Zou, S Singh, Y Nian, J He, J Cai, S Mansour, H Su, "Glean: Generalized Category Discovery with Diverse and Quality-Enhanced LLM Feedback". *Proceedings of the 17th European Chapter of the Association for Computational Linguistics (EACL)*, 2026.

[2] I Shalyminov, H Su, JW Vincent, S Singh, J Cai, J Gung, R Shu, et al., "Controllable Conversational Theme Detection Track at DSTC 12". *Proceedings of the Twelfth Dialog System Technology Challenge, 74–90*, 2025.

[3] Y Zhang, S Singh, S Sengupta, I Shalyminov, H Su, H Song, S Mansour, "Can Your Model Tell a Negation from an Implicature? Unravelling Challenges with Intent Encoders". *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.

[4] H Aboutalebi, H Song, Y Xie, A Gupta, L Sun, H Su, I Shalyminov, et al., "MAGID: An Automated Pipeline for Generating Synthetic Multi-Modal Datasets". *Proceedings of the 2024 Conference of the North American Chapter of the ACL (NAACL)*, 2024.

[5] L Tang, I Shalyminov, A Wong, J Burnsky, J Vincent, Y Yang, S Singh, et al., "TofuEval: Evaluating Hallucinations of LLMs on Topic-Focused Dialogue Summarization". *Proceedings of the 2024 Conference of the North American Chapter of the ACL (NAACL)*, 2024.

[6] H Song, I Shalyminov, H Su, S Singh, K Yao, S Mansour, "Enhancing Abstractiveness of Summarization Models through Calibrated Distillation". *Findings of the Association for Computational Linguistics: EMNLP 2023, 7026–7036*, 2023.

[7] KH Huang, S Singh, X Ma, W Xiao, F Nan, N Dingwall, WY Wang, et al., "SWING: Balancing Coverage and Faithfulness for Dialogue Summarization". *Findings of the Association for Computational Linguistics: EACL 2023, 512–525*, 2023.

[8] S Singh, AF Aji, GS Tomar, C Christodoulopoulos, "A Relation Extraction Dataset for Knowledge Extraction from Web Tables".
Proceedings of the 29th International Conference on Computational Linguistics (COLING), 2022.

[9] Y Zhou, S Singh, C Christodoulopoulos, "Tabular Data Concept Type Detection Using Star-Transformers". *Proceedings of the 30th ACM International Conference on Information and Knowledge Management (CIKM)*, 2021.

[10] G Singh, S Singh, J Wong, A Saffari, "Relation Extraction from Tables Using Artificially Generated Metadata". *arXiv preprint arXiv:2108.10750*, 2021.

[11] S Singh, HM Hang, "A Continuous Learning System for Face Clustering and Recognition". *2021 IEEE International Conference on Consumer Electronics (ICCE), 1–6*, 2021.

[12] S Singh, K Mittal, "Internet of Everything Smart Environment". *2015 International Conference on Green Computing and Internet of Things (ICGCIoT)*, 2015.

MENTORED STUDENTS

- Alexa AI: Ruxandra-Teodora Fleseriu, Abbie Sims
- AWS AI: Sam Mayers, Steeve Huang, Yuwei Zhang, Henry Peng Zou, Gowtham Premananth

INVITED TALKS

- *"Making Alexa More Knowledgeable"* — AI & Big Data Expo London 2022, Olympia London.
- *"Making Alexa More Knowledgeable"* — REWORK London AI Summit 2022.